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A Database Selection Expert System Based on Reference Librarian's Database Selection Strategy: A Usability and Empirical Evaluation

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The proliferation of digital information resources and electronic databases challenges libraries and demands that libraries develop new mechanisms to facilitate and better inform user selection of electronic databases and search tools. We developed a prototype, Web-based database selection expert system based on reference librarian's database selection strategy. This system allows users to simultaneously search all databases available to identify databases most relevant to their quests using free-text keywords or phrases taken directly from their research topics. This article reports on (1) the initial usability test and evaluation of the Selector—the test design, methodology used, performance results; (2) summary of search query analyses; (3) user satisfaction measures; (4) the use of the findings for further modification of the Selector; and (5) the findings of using randomly selected subjects to perform a usability test with predefined searching scenarios. Future prospects of this research have also been discussed in the article.

Introduction

The past decade has seen an explosion in the numbers and types of electronic information resources available to identify and retrieve articles and other information relevant to a scholar's research needs. Library users are confronted with a myriad of online, CD-ROM bibliographic and full-text databases, and other digital information resources. Past practices used by academic libraries—i.e., those that rely on in-person consultation with librarians or that expect end-users to select from simple menus of available resources—are not enough. Improvements are needed in the way libraries facilitate the on-line database selection process, especially for the increasing number of end-users who access library resources remotely.

Academic libraries are trying a variety of strategies of varying sophistication to address this need. Our effort has focused on analyzing and then emulating the techniques most commonly used by reference librarians when helping library users identify databases to search. We developed a database selection algorithm, which mimics the database selection strategy used by reference librarians (Ma, 2000). We gathered detailed characterization data describing each of 146 databases currently available to the University of Illinois at Urbana-Champaign (UIUC) Library. These data were put into a SQL database. We acquired database-controlled vocabularies and enter into this SQL database to further characterize 82 of the 146 databases. Due to time constraints, we were not able to get the controlled vocabulary files for all of the databases included. 46 of the 82 databases had full controlled vocabulary files from database producers. The remaining 36 of 82 databases contained only 5–10% of data-mining subject terms derived from 1998's database records (Ma, 2000). We developed a prototype Web-based electronic database selection expert system, named Smart Database Selector (the Selector), in 1999 (Ma, 2000). The target users of the Selector are general library users and library public service staff who assist users for their information needs.

The query interface includes three search forms (see Fig. 1):

Form 1 is the search form for searching free-text keywords and phrases with Boolean operators. The free-text keywords can be word(s) or phrase(s) from the topic, can be subject terms, names (last name first), geographic areas, etc. This search form also automatically handles right-hand truncations (e.g., *aeroengin* will return databases that index "aerospace engineering," "aeronautical engineering," "aerospace engineers," etc.).

Form 2 allows a user to choose a subject term, if he/she has difficulty expressing the topic in free-text terms. The Selector will display a list of databases relevant to the selected subject term.

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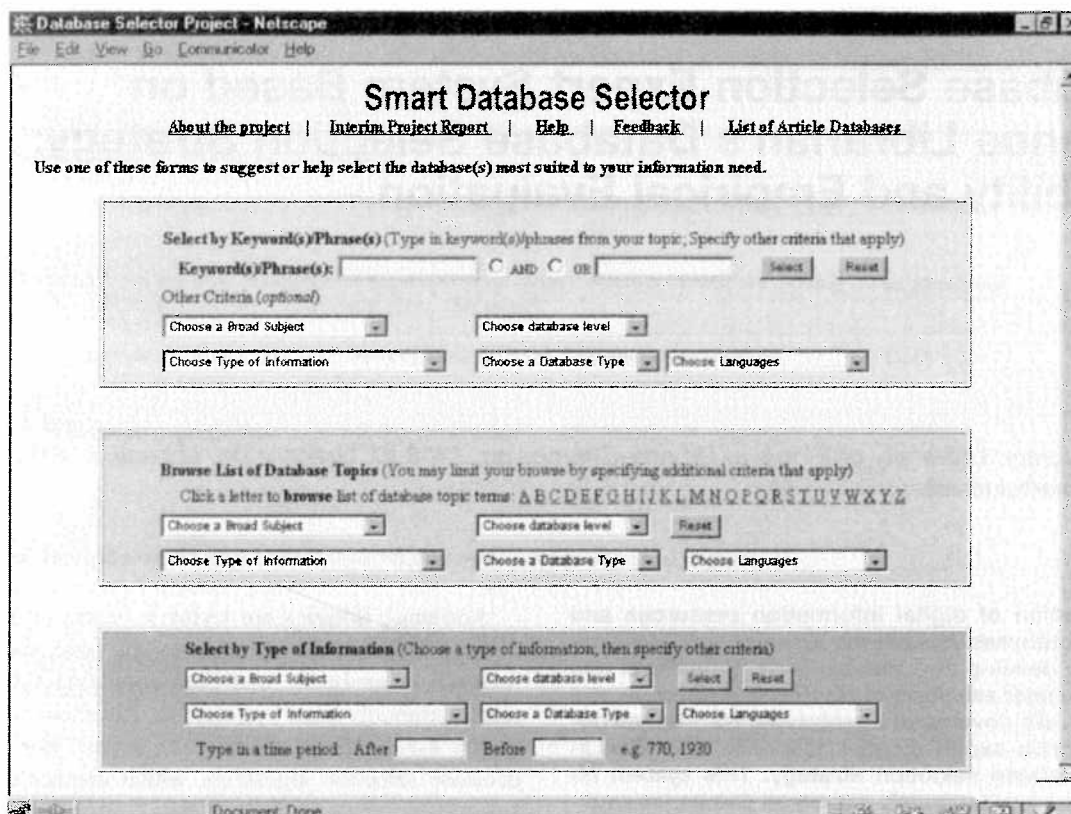


FIG. 1. The tested interface of the Smart Database Selector.

Note:

"Choosing a Broad Subject" includes these options:

Arts & Humanities
Life Sciences and Medicine
Physical Sciences & Engineering
Social Sciences
Business and Economics
Education
Law
Multisubjects

"Choose Type of Information" includes these options:

A-V review	Letters to editor
Biographical information	Literary works, literature criticism/review
Book chapters	Magazine articles
Book review	Manuscripts
Conference proceedings	Maps
Current events	Music review
Dissertations & theses	Newsletters
Editorials	Newspaper articles
Exhibit review	Organizations/Associations
Film review	Pamphlets
Government documents	Patents
Grant information	Photography
Historical events	Readers' advisory resource
Historical information	Research reports
Illustrations/portraits	Software review
International (Other Countries)	Speeches
Information	Statistical information
Interviews	Subject review articles
Journal articles	Trade publications
Legislative information	Treaties

"Choose a Database Level" includes these options:

General interests level
Scholarly level for humanities
Scholarly level for social sciences
Scholarly level for scientific/technical information

"Choose a Database Type" includes these options:

Bibliographic
Directory
Full Text
Numerical

"Choose Languages" includes these options:

English
English & Other languages
Both

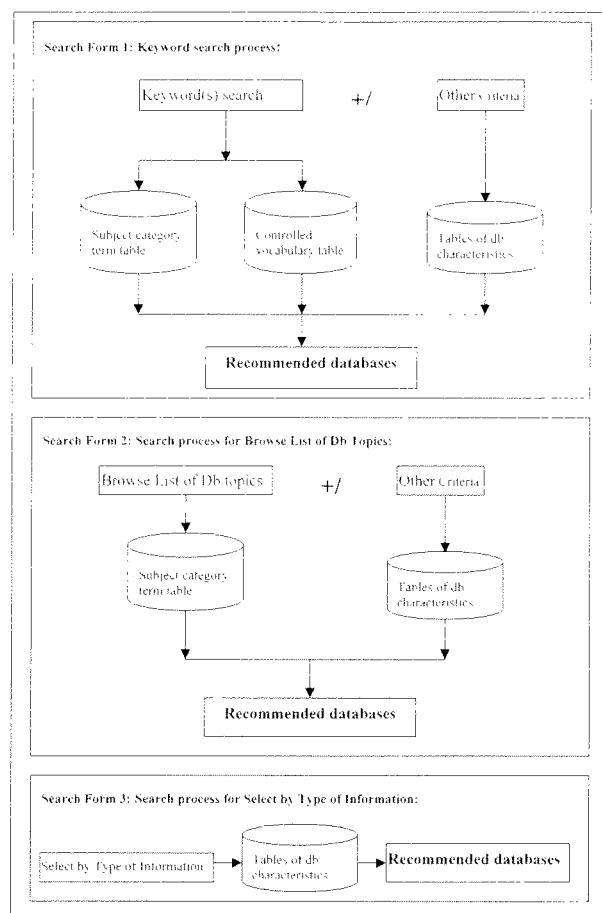


FIG. 2. Search processes of individual search forms and searching form data structure.

Form 3 allows a user to choose databases based on database characteristics from different picklists, such as "Choose a Broad Subject," "Choose Type of Information," "Choose a Database Level," "Choose a Database Type."

All three forms allow users to simultaneously search all of the 146 databases included and recommend databases most relevant to the search request. All three forms provide optional criteria to allow users to narrow their searches to a specific subject area, type of information desired, academic level of the databases, database type (such as full-text, statistical, etc.) and language. The three search forms work separately and can be maintained and updated separately based on needs/usage interests. Figure 2 indicates the indices and the logic behind each form.

This article reports on initial usability testing and evaluation of the Selector. We describe the test design, methodology used, and performance results. In addition to reporting recall and precision measures, we summarize search query analyses and report on user satisfaction measures, and the use of the findings for further modifications of the Selector interface and configuration. We also report on the findings of using randomly selected subjects to perform a usability test with predefined searching scenarios. Future prospects of this research have also been discussed in this article.

Literature Review

This literature review is primarily focusing on the evaluation of expert systems developed for database selection. Literature reviews reveal that though many expert systems designed to help users with database selection have been developed, few evaluations have been completed describing the performance and success of these systems (Lancaster, 1997, Lancaster & Smith, 1997, Lancaster, Ulvila, Humphrey, Smith, Allen, & Herner, 1996).

Hu (1988) reported an evaluation of the InfoMaster (a version of EASYNET by Telebase System Inc.), which was a menu-driven choice selection system. He measured the InfoMaster's performance by comparing the search results and the consistency in database selection between searches made by four experienced on-line reference librarians and those done by inexperienced searchers aided by InfoMaster. The evaluation did not use recall and precision ratios to measure effectiveness of the system performance. The result of the study showed that database selection by InfoMaster still heavily depended on the accuracy of the human decisions in their selection of the appropriate subject field. InfoMaster was unable to suggest useful selection if the initial subject determination was incorrect. However, InfoMaster could select databases as well as experienced librarians when the user properly selected the subject area. This indicated that InfoMaster relied upon users' ability to translate their topics into InfoMaster's existing subject areas. If a user was not able to convert his/her topic into the existing predefined subject areas or the topic fell out of the existing predefined subject areas, InfoMaster was not useful.

Morris (1994) reported an evaluation of CIDA, a menu-driven expert system tool for the selection of databases in the business area, including information on companies within the United Kingdom. The evaluation compared the performance of CIDA to specific printed database directories of business databases (Directory of International Sources of Business Information and the Online/CD-ROM Business Sourcebook) and to TOM (The Online Manual on Disk). This evaluation did not use recall and precision measurement to indicate the evaluated system's performance either. Two groups of 12 library school students were recruited to use the tools to recommend business-related databases. The search results were then assessed by an on-line information retrieval expert to compare the accuracy and consistency of the database recommendations. The evaluation results showed that CIDA was significantly more accurate and consistent than either the printed database guides or TOM. Subjects showed confidence and preference for CIDA. However, the System also relied on users' ability to convert their research topics into CIDA's existing predefined subject areas. If a user failed to do it properly, the system failed.

Many studies, however, were done to evaluate the efficiency and effectiveness of information retrieval systems using recall and precision. Although often criticized, recall and precision are sufficient to determine the system's per-

formance (Lesk & Salton, 1968; Rorvig, 2000). Lancaster (1969) reported an evaluation of the MEDLARS service system for medical literature analysis and retrieval at the National Library of Medicine. The evaluation utilized recall and precision measures to indicate the System's performance. But, MEDLARS was not an end-user searching system. Users submitted search requests, librarians administered the searches and then sent search results to users for "recall" and "precision" analysis (Lancaster, 1969). The evaluation results revealed that MEDLARS systems' overall precision ratio was 50.4%. Precision ratio based on major value articles was 25.7%. Overall recall ratio was 57.7%. Recall ratio based on major value articles was 65.2%. However, in this evaluation, the recall ratio was estimated based on retrieval performance in relation to a number of documents, judged relevant by the requester, but found by means outside MEDLARS.

Adelman, Ulvila, and Lehner. (1990) described a method for testing key aspects of knowledge-based systems. Lancaster et al. (1996) interpreted and further refined the evaluation method of knowledge-based systems introduced by Adelman. Lancaster et al. recommended that systems designed for expert users be evaluated on different criteria than systems designed for novices, and suggested different methods to test against the criteria.

Based on the Database Selector system that we have developed, taking into account the methods that Adelman and Lancaster described, we designed evaluation criteria that would meet our special environment and needs.

Objectives of the Evaluation

Usability Testing Scope & Objectives

Our evaluation only focused on search Form 1—Keyword Search. Search Forms 2 and 3 were not evaluated in the usability testing described here, because Search Form 2 and 3 were designed using different methodology and serve different users needs. Search Form 2 and 3 will be discussed in subsequent articles. The testing of Search Form 1 was intended to satisfy the following objectives:

- (1) To measure the Selector performance in suggesting electronic database resources relevant to users' queries. This objective determined the analysis performed: (a) to calculate the average (mean) of per-search recall and precision ratios to better quantify how well the Selector identifies databases most relevant to users' queries. (b) To quantify the effectiveness of database-controlled vocabulary as a means of characterizing database content. Our initial hypothesis was that inclusion of database-controlled vocabulary would facilitate discovery of relevant databases without adversely impacting the precision of database selection process. To test this hypothesis we calculated average per-search recall and precision ratios for three sets of databases included in the Selector: (i) the complete set of databases included in the Selector; (ii) the subset of the databases in the

Selector described using database thesaurus and data-mining subject terms derived from 1998's database records in addition to librarian-assigned subject category terms; (iii) the subset of the databases in the Selector described using only librarian-assigned subject category terms. (c) To measure the recall and precision ratios of searches which combine keyword search with the "Other Criteria" options (see Note for Fig. 1) to see if the "Other Criteria" limiting options help improve the search quality as expected: (i) searches using "Other Criteria" options to help limit the search; (ii) searches not using "Other Criteria" options.

- (2) To discover how users, both library professionals and library users (frequent and nonfrequent), utilize the Selector, and to find out if search strategies and query formulation make any difference on the scale of recommending relevant databases. This objective determined the analysis performed: (a) comparing the different searching styles of users of the Selector to see if users with library experience perform searches with better recall and precision ratios than users without library experience: (i) frequent library users versus nonfrequent library users; (ii) library users versus library staff.
- (3) To identify ways in which the Selector can be improved. This objective determined the analysis performed: (a) analyzing failed searches—searches that did not retrieve any results and searches that retrieved irrelevant databases—to identify ways for further improvements.
- (4) To solicit user reactions to the usefulness of the Selector. This objective determined the actions performed: (a) collecting and analyzing the posttest questionnaires; (b) conducting brief interviews at the end of each test session.

Methodology

Data collection during usability testing was done using a combination of transaction logs and surveys filled out by test subjects. Test administrator also observed subjects while they used the Selector to perform specific tasks and briefly interviewed subjects at conclusion of the test sessions.

Usability Tests

User Population. Two groups of usability test subjects were recruited—a library user group (undergraduates, graduates, and teaching faculty) and a library staff group (both librarians and staff who had at least 1 year of library public service experience). Table 1 describes the populations used in the tests.

We solicited subjects through e-mail announcements, postings in student dormitories and academic office buildings, and class announcements. We received over 100 UIUC affiliated respondents. Test subjects were selected to represent different academic levels and fields. The library user group included students majoring in humanities, social sciences, science, and engineering, and representing all lev-

TABLE 1. Population.

Usability test groups	Population sampled	Numbers of subjects	
		#	%
Library user group	Graduates/faculty	8	36.4
	Undergraduates	8	36.4
Library staff group	Librarians and staff	6	27.3
Total		22	100%

els from freshmen to senior, graduate students, and teaching faculty. This group also contained American-born and foreign-born individuals. The library staff group included professional librarians, full-time library staff and students from the Graduate School of Library Information Science (GS-LIS) who had received at least 1 year of professional library training and had at least 1 year of library working experience (e.g., as a library graduate assistant). Because of the budget constraints, we were only able to recruit 22 subjects.

Search scenarios for testing were selected from real reference questions collected at the Library's central reference desk, with an intention to represent a variety of subject areas. To find out if the system was useful for library users and for library professionals as well, we decided to give library user group and library staff group different sets of search scenarios to perform. The Library staff group's search scenarios included a few tougher reference questions and questions that had more conditions. Figures 3 and 4 show the search scenarios used in this usability test.

Usability Test Procedure. Usability testing took place from mid-April to early June of 2000. The testing was scheduled in groups consisting of one to four individuals based on the availability of the subjects' schedules. Each session took approximately 1 to 2 hours or until the subjects finished all the tasks required. Before beginning, subjects were told the purpose of testing and the general outlines of what they were expected to do. No specific instructions or training were given on how to use the Selector tool (a brief on-line help page was available). No discussion or talking among subjects was allowed during the test. Subjects were required

Questions for library user group:

- 1) Need articles on U.S. invasion of Normandy.
- 2) What is being done to deal with aging nuclear power plants? (need to find full-text articles about this topic for an undergraduate presentation)
- 3) Need to find articles and statistics on the current trend of women studying computer science.
- 4) Need to find articles that compare different literary reviews and criticism on Wilde, Oscar's "The Importance of being Earnest."
- 5) Water quality in developing countries (need to do in depth research as for a research project presentation)
- 6) "Rosenkrantz and Guilderstern are Dead" (want articles which compared the play and the film)
- 7) A comparative look at the role of the church in the Mexican American community (need to do in depth research for a graduate dissertation)
- 8) Alcohol abuse in European countries (need to find full-text articles about this topic quickly for an undergraduate term paper assignment)
- 9) Want articles which outline the current events on stock market investment for a short undergraduate presentation
- 10) Gun control in the United States (need to find full-text articles about this topic for an undergraduate term paper)

FIG. 3. Search scenarios for library user group.

Questions for Library staff group:

- 1) Why do people buy a certain type of shoes? Do they think the shoes they decide upon represent something about their personality? (The user needed primary and scholarly information to write a market report about consumer behavior on buying footwear)
- 2) What is being done to deal with aging nuclear power plants? (The user needed general level information to write a 6 page term paper. Full-text was preferred)
- 3) A patron wanted articles which outline the current events on stock market investment
- 4) Who was Lady Baltimore? (A user was researching on Latin American culture and would like to find out why many restaurants are named after her name)
- 5) A user was looking for federal and non-federal grants and funding opportunities to support his research in the area of environmental science
- 6) A student needed to compare different literary reviews and criticism on Oscar Wilde's "The Importance of being Earnest" for his class assignment
- 7) A student was looking for scholarly information about water quality in developing countries for a term paper project
- 8) "Rosenkrantz and Guilderstern are Dead" The patron wanted the articles which compared the play and the film
- 9) A student taking History 263 was looking for information on the topic "A comparative look at the role of the church in the Mexican American community" for her 10 page term paper project
- 10) A user was looking for databases which include literary reviews or criticism of children's literature in the area of arts and humanities
- 11) A user was looking for most recent statistical information about Arab American population in the U.S.
- 12) Current trends in women studying computer science. The patron needed most recent statistical figures and articles to write a report

FIG. 4. Search scenarios for library staff group.

to answer the predefined search questions in the same order as listed in Figures 3 and 4.

After completing initial demographic background questionnaires, subjects were asked to pick a topic of their own choosing and use the Selector to suggest databases. By so doing we gave the subjects the opportunity to get familiar with the Selector. Then, they were asked to use the Selector to identify electronic resources for the predefined search questions. The entire test process was carefully observed. At the end of the test session all individuals were asked to complete a Post-Test Questionnaire providing qualitative feedback on the usability of the interface. Before leaving, a brief interview was typically conducted—for example, asking if she/he felt the Selector was useful, if he/she liked the Selector, if he/she had suggestions for improvements, etc. Transaction logs were kept of all searches done by subjects during the testing. Full details regarding search arguments submitted, search limits applied, and results retrieved (i.e., lists of database names) were recorded in transaction logs for later analysis.

Data Analysis. The Selector's system performance was measured using the standard ways of measuring search results: Recall and Precision. Recall is defined as the proportion of relevant material retrieved, while Precision is the proportion of retrieved material that is relevant. In other words, Recall measures the proportion of the relevant items in the collection that were actually retrieved by the system, and is therefore a measure of the efficiency of the retrieval ability of the system. Precision, on the other hand, takes the total number of documents retrieved, and measures the percentage of those retrieved documents that are actually relevant. It is, therefore, a measure of how accurate the system is in recognizing and retrieving only relevant documents.

$$\text{Recall} = \frac{\text{Number of items retrieved and relevant}}{\text{Total relevant in collection}} \times 100$$

$$\text{Precision} = \frac{\text{Number of items retrieved and relevant}}{\text{Total retrieved}} \times 100$$

In averaging recall and precision measures over test group populations, standard deviation was calculated to indicate the variability in these measures user to user and search to search. For these analyses, standard deviation was calculated as:

$$\text{Standard deviation} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

where N is the number of samples used to calculate the average recall or precision, X_i is each recall or precision value, and $\bar{X}$ is the calculated average recall or precision value.

Theoretically, higher recall and precision measures imply a better, more successful search. Smaller standard deviations imply greater consistency in search success across the user population. However, according to Cleverdon (1974), it is "... only in very exceptional circumstances that a user does require a 100% recall of documents ... For many subjects, a recall ratio of 25% or less of the relevant document will give a complete 100% recall of information." He gave an example, "Only 96 documents were necessary to have every item of information that was contained in a complete collection of over 4,000 papers on the applications of milkweed." It is also important to point out that a successful search depends not only on the tested system's performance, but also on a user's understanding of the topics, search strategy, query construction, and proper searching behavior.

Result Analysis

System Performance

Usability test subjects did a total of 945 searches. Of these, 672 keyword searches (using search Form 1 of the interface) were done to recommend databases in an effort to answer one of the predefined search questions. The rest of the searches used Forms 2 or 3 (which are not included in this evaluation) or were done to answer a question of a subject's own devising. Of the 672 keyword searches analyzed, 457 retrieved at least one database name. The rest of the 672 (215 out of 672) keyword searches retrieved 0 hits. Recall and precision measures were calculated for each of these 457 searches, which included searches that recommended irrelevant databases (0 recalls) and searches that recommended too many databases. All failed searches, including searches that retrieved no databases (0 hit), searches that retrieved irrelevant databases (0 recalls), and searches that retrieved too many hits were analyzed individually to determine the most common reasons for search failures.

To do this, the author first made a preliminary determination of which all of the 146 information resource databases characterized in the Selector tool might be useful for answering each of the predefined search questions. Each question-specific list of relevant resource databases was then verified (narrowed) by actually using identified resource databases to try and answer the predefined search question and then assessing the degree of usefulness of each resource database for the question.

Having thus identified a list of relevant resource databases for each predefined search question, recall and precision quality measures were calculated. The results indicated that recall and precision quality varied significantly question to question, user to user, and search to search. To aggregate the results for this article, recall and precision averages (arithmetic mean) and standard deviations from those averages were also calculated. A number of trends are clear from inspection of those results.

Per-search recall and precision measures calculated for searches done to suggest databases to answer the predefined search questions were averaged on a per question basis over different user groups. Table 2 shows per question average recall and precision measures for searches done by library user group subjects (a total of 297 searches that retrieved databases). Averages for library staff subjects (who were given a different set of predefined search questions) are shown in Table 3 (a total of 160 searches that retrieved databases).

As mentioned above, controlled vocabulary terms and partial subject terms from databases were acquired for a total of 82 of the 146 resource databases included in the Selector. The rest of the 146 databases in the Selector were described according to the characteristics of the databases (Ma, 2000)—broad subject areas, contents and formats, academic levels, and data types—and subject category areas covered. While searching, keyword search strings submitted to the Selector tool were searched both against these 82 controlled vocabularies as well as against the indices of database characteristics and subject category terms. Because one of the premises of this study was that free-text keyword searches done against indices of database-controlled vocabularies would facilitate identification of resource databases useful for meeting a particular information need, separate recall and precision averages were calculated according to availability of controlled vocabulary terms.

Examining Tables 2 and 3, it is clear that (as anticipated) recall from free-text keyword searches done by both groups is much better when such searches can be done against database-controlled vocabularies than when such searches are done only against database characteristics and subject category terms generated by librarians. Almost none of those databases for which we did not include controlled vocabulary terms in the Selector's index were discovered via end-user free-text keyword searches, even though a number of these databases were judged relevant to a particular question. Average per question precision varied from a high of 94.0% to a low of 10%, and was significantly

TABLE 2. Per question recall and precision, library user group.

Average recall (R) and precision (P), $\pm$ standard deviations, calculated considering:			
Question	All databases	Only databases for which controlled vocabulary terms or partial subject terms were available	Only databases for which controlled vocabulary terms were not available
1	R: 26.8% ($\pm 11.7\%$) P: 74.8% ($\pm 28.9\%$)	R: 49.1% ($\pm 21.4\%$) P: 74.8% ($\pm 28.9\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
2	R: 44.4% ($\pm 26.3\%$) P: 55.0% ($\pm 35.1\%$)	R: 49.4% ($\pm 29.3\%$) P: 55.9% ($\pm 35.1\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
3	R: 33.8% ($\pm 20.1\%$) P: 72.5% ($\pm 20.7\%$)	R: 44.0% ($\pm 27.9\%$) P: 69.0% ($\pm 27.4\%$)	R: 3.3% ($\pm 6.8\%$) P: 10.3% ($\pm 24.2\%$)
4	R: 9.6% ($\pm 11.6\%$) P: 30.4% ($\pm 37.8\%$)	R: 14.5% ($\pm 17.4\%$) P: 30.4% ($\pm 37.8\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
5	R: 28.8% ($\pm 24.4\%$) P: 44.4% ($\pm 28.1\%$)	R: 43.1% ($\pm 36.6\%$) P: 44.4% ($\pm 28.1\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
6	R: 48.1% ($\pm 9.8\%$) P: 79.0% ($\pm 33.8\%$)	R: 48.1% ($\pm 9.8\%$) P: 79.0% ($\pm 33.8\%$)	N/A
7	R: 20.0% ($\pm 18.9\%$) P: 10.0% ($\pm 11.6\%$)	R: 33.3% ($\pm 31.6\%$) P: 10.0% ($\pm 11.6\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
8	R: 34.7% ($\pm 23.5\%$) P: 35.6% ($\pm 31.0\%$)	R: 52.0% ($\pm 35.3\%$) P: 35.6% ($\pm 31.0\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
9	R: 59.2% ($\pm 21.9\%$) P: 50.7% ($\pm 24.8\%$)	R: 70.0% ($\pm 25.5\%$) P: 50.7% ($\pm 24.7\%$)	R: 5.0% ($\pm 22.4\%$) P: 5.0% ($\pm 22.4\%$)
10	R: 34.8% ($\pm 15.8\%$) P: 57.1% ($\pm 36.6\%$)	R: 60.8% ($\pm 27.6\%$) P: 57.1% ($\pm 36.6\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)

different with databases for which controlled vocabularies were included and those for which controlled vocabulary terms were not available.

Six of the predefined questions asked of library user group subjects were also asked of library staff subjects (although not in the same order). Library staff group recall

TABLE 3. Per question recall and precision, library staff group.

Average recall (R) and precision (P), $\pm$ standard deviations, calculated considering:			
Question	All databases	Only databases for which controlled vocabulary terms or partial subject terms were available	Only databases for which controlled vocabulary terms were not available
1	R: 39.3% ($\pm 32.1\%$) P: 28.9% ($\pm 28.2\%$)	R: 49.1% ($\pm 40.1\%$) P: 28.9% ($\pm 28.2\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
2	R: 44.6% ($\pm 20.7\%$) P: 81.4% ($\pm 29.6\%$)	R: 55.6% ($\pm 23.0\%$) P: 81.4% ($\pm 29.6\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
3	R: 58.3% ($\pm 16.2\%$) P: 71.9% ($\pm 18.6\%$)	R: 70.0% ($\pm 19.4\%$) P: 71.9% ($\pm 18.6\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
4	R: 53.3% ($\pm 17.2\%$) P: 94.0% ($\pm 19.0\%$)	R: 53.3% ($\pm 17.2\%$) P: 94.0% ($\pm 19.0\%$)	N/A
5	R: 77.8% ($\pm 44.1\%$) P: 77.8% ($\pm 44.1\%$)	R: 77.8% ($\pm 44.1\%$) P: 77.8% ($\pm 44.1\%$)	N/A
6	R: 20.9% ($\pm 18.8\%$) P: 46.1% ($\pm 37.4\%$)	R: 33.4% ($\pm 24.8\%$) P: 45.3% ($\pm 38.4\%$)	R: 5.9% ($\pm 17.6\%$) P: 7.1% ($\pm 24.4\%$)
7	R: 44.4% ($\pm 20.3\%$) P: 67.6% ($\pm 22.0\%$)	R: 66.7% ($\pm 30.5\%$) P: 67.6% ($\pm 22.0\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
8	R: 54.3% ($\pm 14.4\%$) P: 38.5% ($\pm 25.3\%$)	R: 54.3% ($\pm 14.4\%$) P: 38.6% ($\pm 25.2\%$)	N/A
9	R: 32.0% ($\pm 14.0\%$) P: 12.1% ($\pm 6.6\%$)	R: 53.3% ($\pm 23.3\%$) P: 12.1% ($\pm 6.6\%$)	R: 0.0% ($\pm 0.0\%$) P: 0.0% ($\pm 0.0\%$)
10	R: 53.7% ($\pm 26.1\%$) P: 43.2% ($\pm 22.3\%$)	R: 62.2% ($\pm 27.3\%$) P: 44.3% ($\pm 21.5\%$)	R: 11.1% ($\pm 33.3\%$) P: 1.1% ($\pm 3.3\%$)
11	R: 62.5% ($\pm 23.1\%$) P: 22.4% ($\pm 15.7\%$)	R: 62.5% ($\pm 23.1\%$) P: 22.5% ($\pm 15.6\%$)	N/A
12	R: 30.6% ($\pm 11.8\%$) P: 47.2% ($\pm 25.4\%$)	R: 39.5% ($\pm 14.8\%$) P: 55.9% ($\pm 25.7\%$)	R: 3.7% ($\pm 11.1\%$) P: 5.6% ($\pm 16.7\%$)

TABLE 4. Per question recall and precision searches with/without limits, library user group.

Average recall (R) and precision (P) $\pm$ standard deviation, calculated considering:			
Question	All keyword searches	Keyword searches in conjunction with limits	Keyword searches only
1	R: 26.8% ($\pm$ 11.7%) P: 74.8% ($\pm$ 28.9%)	R: 18.9% ($\pm$ 9.4%) P: 75.6% ($\pm$ 30.7%)	R: 31.2% ($\pm$ 10.6%) P: 74.3% ($\pm$ 28.6%)
2	R: 44.4% ($\pm$ 26.3%) P: 55.9% ($\pm$ 35.1%)	R: 34.3% ($\pm$ 25.6%) P: 69.9% ($\pm$ 36.9%)	R: 62.3% ($\pm$ 16.9%) P: 31.0% ($\pm$ 6.7%)
3	R: 33.8% ($\pm$ 20.1%) P: 72.5% ($\pm$ 20.7%)	R: 18.4% ($\pm$ 11.7%) P: 75.2% ($\pm$ 28.0%)	R: 48.1% ($\pm$ 14.9%) P: 70.0% ($\pm$ 11.1%)
4	R: 9.6% ($\pm$ 11.6%) P: 30.4% ($\pm$ 37.8%)	R: 5.9% ($\pm$ 8.0%) P: 26.7% ($\pm$ 38.9%)	R: 29.6% ($\pm$ 5.7%) P: 50.0% ($\pm$ 25.0%)
5	R: 28.8% ($\pm$ 24.4%) P: 44.4% ($\pm$ 28.1%)	R: 20.1% ($\pm$ 17.8%) P: 44.0% ($\pm$ 30.3%)	R: 50.0% ($\pm$ 26.4%) P: 45.2% ($\pm$ 23.5%)
6	R: 48.1% ($\pm$ 9.8%) P: 79.0% ($\pm$ 33.8%)	R: 46.9% ($\pm$ 12.5%) P: 79.5% ($\pm$ 37.6%)	R: 50.0% ($\pm$ 0.0%) P: 78.3% ($\pm$ 28.4%)
7	R: 20.0% ($\pm$ 18.9%) P: 10.0% ($\pm$ 11.6%)	R: 11.8% ($\pm$ 14.7%) P: 10.0% ($\pm$ 13.7%)	R: 42.5% ($\pm$ 7.1%) P: 10.0% ($\pm$ 1.0%)
8	R: 34.7% ($\pm$ 23.5%) P: 35.6% ($\pm$ 31.0%)	R: 29.6% ($\pm$ 25.9%) P: 40.7% ($\pm$ 35.3%)	R: 47.6% ($\pm$ 6.3%) P: 22.6% ($\pm$ 6.4%)
9	R: 59.2% ($\pm$ 21.9%) P: 50.7% ($\pm$ 24.8%)	R: 61.5% ($\pm$ 19.0%) P: 57.2% ($\pm$ 22.3%)	R: 50.0% ($\pm$ 33.3%) P: 24.8% ($\pm$ 17.2%)
10	R: 34.8% ($\pm$ 15.8%) P: 57.1% ($\pm$ 36.6%)	R: 32.3% ($\pm$ 17.3%) P: 67.0% ($\pm$ 36.5%)	R: 42.9% ($\pm$ 0.0%) P: 24.9% ($\pm$ 2.2%)

was better on the six questions in common. Library staff precision measures were better in five out of the six cases than Library user group. These results suggest that library staff did tend to formulate "better" search queries when using the Selector tool.

An analysis similar to the above was done to see if there were any notable differences between searching effectiveness of graduate/faculty usability test subjects compared to undergraduate student usability test subjects. No significant differences were found for our sample user groups.

Tables 4 and 5 show the impact on performance of combining keyword searches with "other criteria" options. We expected that the use of optional "other criteria" would result in better precision but also some loss of recall. While the general trends were as expected, the magnitude of the effects on recall and precision were not as expected. Optional criteria were used on 197 out of the 297 keyword searches by library user group subjects that retrieved databases. Library user group subjects seemed to have a difficult time using the optional criteria effectively on some questions (in some instances recall was lowered dramatically when optional criteria were applied though precision increased somewhat), suggesting a need to rework the criteria options presented (which has now been done). Library staff subjects used the optional criteria a greater percentage of the time (200 out of 297) and seemed able to use them more effectively (see Table 5).

Subjects were allowed to submit multiple searches in an effort to identify best resource databases for a particular question. Most users did so. The assumed advantage of this strategy was to give users the opportunity to revise search queries and to resubmit searches until they were satisfied

with the databases recommended. To see if this strategy indeed helped users discover more relevant databases, an "average recall" was calculated across all the searches done by each participant for each question. The results of this analysis are shown in Tables 6 and 7. Effective "per user" recall was calculated by combining all sets retrieved by a given user from all searches by that user regarding a particular question and then calculating the average percentage of the question-specific relevant databases that were recommended in the combined superset. The results show that effective recall averages per user (combining all sets of searches) were generally higher than per search recall averages presented in Tables 2 and 3. Tables 6–7 also confirm that Library staff recalls (however calculated) were better than library user group recalls. However, other factors such as native language and frequency of library uses were not major factors in per-user recall measures.

Summary of Failed Searches and Possible Solutions

As mentioned above, there were 672 keyword searches done by usability test subjects to identify databases for the predefined search questions. Of the 672 searches, 215 searches did not retrieve any databases (0 hit searches), 64 of the 672 retrieved databases, but per-search recall was 0 (databases recommended were not relevant), 9 of the 672 keyword searches retrieved more than 25 databases (too many hit searches). We defined these searches as failed searches. Table 8 shows the failed searches by different test group subjects.

Every failed search was analyzed in detail for the various factors contributing to the problems, and then classified into

TABLE 5. Per question recall and precision searches with/without limits, library staff group.

Question	Average recall (R) and precision (P) ($\pm$ standard deviation) calculated considering:		
	All keyword searches	Keyword searches in conjunction with limits	Keyword searches only
1	R: 39.3% ($\pm$ 32.1%) P: 28.9% ($\pm$ 28.2%)	R: 70.0% ($\pm$ 28.1%) P: 30.2% ($\pm$ 31.0%)	R: 80.0% ($\pm$ 0.0%) P: 23.0% ($\pm$ 7.6%)
2	R: 44.6% ($\pm$ 20.7%) P: 81.4% ($\pm$ 29.6%)	R: 42.5% ($\pm$ 20.1%) P: 85.7% ($\pm$ 26.3%)	R: 70.0% N/A P: 29.2% N/A
3	R: 58.3% ($\pm$ 16.2%) P: 71.9% ($\pm$ 18.6%)	R: 58.3% ($\pm$ 16.2%) P: 71.9% ($\pm$ 18.6%)	R: N/A P: N/A
4	R: 53.3% ($\pm$ 17.2%) P: 94.0% ($\pm$ 19.0%)	R: 51.9% ($\pm$ 17.6%) P: 93.3% ($\pm$ 20.0%)	R: 66.7% N/A P: 100.0% N/A
5	R: 77.8% ($\pm$ 44.1%) P: 77.8% ($\pm$ 44.1%)	R: 100.0% ($\pm$ 0.0%) P: 100.0% ($\pm$ 0.0%)	R: 0.0% ($\pm$ 0.0%) P: 0.0% ($\pm$ 0.0%)
6	R: 20.9% ($\pm$ 18.8%) P: 46.1% ($\pm$ 37.4%)	R: 21.5% ($\pm$ 19.2%) P: 47.4% ($\pm$ 38.2%)	R: 11.1% N/A P: 25.0% N/A
7	R: 44.4% ($\pm$ 20.3%) P: 67.6% ($\pm$ 22.0%)	R: 26.2% ($\pm$ 14.8%) P: 65.5% ($\pm$ 29.0%)	R: 60.4% ($\pm$ 3.9%) P: 69.4% ($\pm$ 15.4%)
8	R: 54.3% ($\pm$ 14.4%) P: 38.5% ($\pm$ 25.3%)	R: 52.9% ($\pm$ 12.1%) P: 36.9% ($\pm$ 27.9%)	R: 58.3% ($\pm$ 20.4%) P: 43.0% ($\pm$ 17.1%)
9	R: 32.0% ($\pm$ 14.0%) P: 12.1% ($\pm$ 6.6%)	R: 20.0% ($\pm$ 16.3%) P: 14.6% ($\pm$ 10.5%)	R: 40.0% ($\pm$ 0.0%) P: 10.4% ($\pm$ 1.7%)
10	R: 53.7% ($\pm$ 26.1%) P: 43.2% ($\pm$ 22.3%)	R: 53.7% ($\pm$ 26.1%) P: 43.2% ($\pm$ 22.3%)	R: N/A P: N/A
11	R: 62.5% ($\pm$ 23.1%) P: 22.4% ($\pm$ 15.7%)	R: 66.7% ($\pm$ 25.8%) P: 27.2% ($\pm$ 15.2%)	R: 50.0% ($\pm$ 0.0%) P: 8.1% ($\pm$ 4.3%)
12	R: 30.6% ($\pm$ 11.8%) P: 47.2% ($\pm$ 25.4%)	R: 57.1% ($\pm$ 5.9%) P: 48.5% ($\pm$ 26.8%)	R: 58.3% N/A P: 36.8% N/A

error groups to identify the causes of the problems. Some failed searches involved more than one category of error types. These failed searches were counted respectively in the categories in which they were classified. Table 9 shows the breakdown of the failed searches.

"0" Hit Searches

Searches that failed to retrieve any databases were defined as "0 hit" searches. The analyses of the causes of these searches are listed below.

Misspelling. Spelling errors only occurred 31 times in the 215 zero-hits searches. In the cases analyzed, the subjects appeared aware of the errors and corrected them automatically.

Keyword Misused. There were a total of 129 cases of using mismatched keyword inputs to search the Selector, which resulted in 0 hits. Major errors that caused the searches to

TABLE 6. Per user recall for each question, library user group.

Question	Average recall ($\pm$ standard deviation) calculated considering:		
	Average number of searches done	All databases	Only databases for which controlled vocabulary terms or partial subject terms were available
1	3.1	30.7% ($\pm$ 10.3%)	56.2% ($\pm$ 18.8%)
2	2.7	59.4% ($\pm$ 18.8%)	66.0% ($\pm$ 20.9%)
3	2.3	41.7% ($\pm$ 16.7%)	63.7% ($\pm$ 23.0%)
4	3.5	15.6% ($\pm$ 13.6%)	23.3% ($\pm$ 20.3%)
5	2.6	36.2% ($\pm$ 25.9%)	54.3% ($\pm$ 38.9%)
6	2.5	44.8% ($\pm$ 15.5%)	54.8% ($\pm$ 15.5%)
7	2.3	25.0% ($\pm$ 19.7%)	41.7% ($\pm$ 32.8%)
8	1.9	42.7% ($\pm$ 22.4%)	64.1% ($\pm$ 33.6%)
9	1.9	64.3% ($\pm$ 15.5%)	75.7% ($\pm$ 17.5%)
10	2.0	40.2% ($\pm$ 7.6%)	70.3% ($\pm$ 13.4%)

TABLE 7. Per user recall for each question, library staff group.

Question	Average recall ($\pm$ standard deviation) calculated considering:		
	Average number of searches done	All databases	Only databases for which controlled vocabulary terms or partial subject terms were available
1	4.8	56.7% ($\pm$ 28.1%)	70.8% ($\pm$ 35.1%)
2	2.3	56.7% ($\pm$ 19.7%)	63.0% ($\pm$ 21.9%)
3	2.6	60.0% ($\pm$ 8.6%)	72.0% ($\pm$ 10.3%)
4	1.7	61.1% ($\pm$ 13.0%)	65.1% ($\pm$ 13.0%)
5	3.3	100.0% ($\pm$ 0.0%)	100.0% ($\pm$ 0.0%)
6	3.5	38.9% ($\pm$ 27.4%)	50.0% ($\pm$ 26.6%)
7	4.0	45.8% ($\pm$ 13.4%)	68.8% ($\pm$ 20.0%)
8	4.4	58.3% ($\pm$ 19.5%)	68.3% ($\pm$ 19.5%)
9	1.7	33.3% ($\pm$ 15.6%)	55.6% ($\pm$ 25.9%)
10	1.5	58.3% ($\pm$ 28.0%)	66.7% ($\pm$ 28.7%)
11	1.9	60.0% ($\pm$ 21.1%)	65.0% ($\pm$ 21.1%)
12	1.6	33.3% ($\pm$ 13.3%)	49.6% ($\pm$ 17.0%)

TABLE 8. Failed searches logged by user groups.

	Library user group		Library staff group		Total
	#:	%:	#:	%:	
Total number of keyword searches					672 (100%)
Failed searches					
0 hit searches	161	23.9%	54	8.0%	215 (31.2%)
0 recall searches	53	7.9%	12	1.8%	65 (9.7%)
Too many hit searches	8	1.2%	1	0.1%	9 (1.3%)
Total number of failed searches	222	33.0%	67	9.9%	289 (43%)

fail could be grouped into three types: (1) typed in the entire topic as keywords, such as for library user search scenario 5 ("water quality in developing countries"). (2) Put unrelated keywords together as a phrase to search the Selector, such as "statistics women computer science," or "job market women" for library user search scenario 3, or "religion Mexican American" for library user topic 7. (3) Made up new keywords, such as "investing now" or "investing current" for library user search scenario 9. These errors could be the results of test subjects not understanding the search requirement, not reading the instructions on the screen, or playing with the Selector tool to see how it reacted to various ways of keyword inputs. When 0 hits occurred, test subjects were able to correct their own errors, and changed their search strategies until they were satisfied with the search results.

Limiting Criteria Misused. There were 81 cases of using mismatched limiting criteria options by all subjects. The cases were grouped into four categories: (1) used conflicting limiting criteria options, such as choosing "Social Sciences" as the Broad Subject Area while selecting "Scholarly for Arts and Humanities" as the database level. This was one of the most common errors made by library user group; (2) proceeded to search the next topic without clearing the limit criteria options from the previous searches; (3) played with the Selector during the test. For example, one subject (Undergraduate 22) did 11 searches on one question using combinations of all different limiting options available, both relevant and irrelevant; (4) used the "Language" limiting option in unexpected ways. The Language option turned out

to be a significant design problem. Fifty-six of the 81 counts of using mismatched limiting criteria options were related to choosing "Language" limiting option. The "Language" limiting criteria was originally designed to help users distinguish databases that included only English language publications from databases, which contained English and other languages publications. However, many subjects applied this criterion to specify their preference for English language publications. When 0 hit occurred, subjects were able to deselect the option and proceeded with the search.

Search Forms Misused. There were eight failed searches related to not using the search form in a correct manner. These included searches with the keywords only in second keyword input box, leaving the first keyword input box blank; multiple keyword inputs without selecting a Boolean operator, etc. Subjects appeared aware of the errors and corrected them automatically when 0 hit occurred.

"0" Recall Searches

"0" (zero) recall searches were those that retrieved and recommended irrelevant databases to the predefined search questions. Analysis of zero recall searches revealed differences between the library user group and the library staff group.

For Library User Group

Keyword Missed Used. This was a serious problem for the user group. Most of the 46 mismatched keyword inputs were related to (1) using keywords too broad or not related, such as using the word "women" alone, or using the word "education" alone for library user search scenario 3, and "nuclear energy" for library user search scenario 2; (2) not typing in the last name first when doing name searches; (3) unable to come up with proper keywords to match the topics. Library user group's searches for databases to answer search scenarios 4, 5, and 7 were generally not well done. These questions (questions 4, 5, 7) were more for graduate students, required more research skills and a good understanding of the topics. For these scenarios as expressed, most undergraduate users appeared to have problems/difficul-

TABLE 9. Breakdown of the failed searches.

Failed searches	Error types			
	Misspelling	Keyword misused	Limiting criteria misused	Search forms misused
0 Hits	31	129	81	8
0 Recalls	3	46	30	
Too many hits		9		
Total (289)	34	184	111	8

ties identifying the proper keywords to search. For example, for library user search scenario 4, the best keywords derive from the title of the work. However, many of the library user group subjects used the author's name as keyword input, which then resulted in zero recall. Most test subjects were not aware of the error until they accessed the suggested databases for relevant citations. This type of errors seemed more associated with users' library research skills and database searching habits.

Limiting Criteria Misused. This is another major problem that caused zero recall searches. There were four major causes for this problem: (1) selecting all limiting criteria options available on the search form, whether appropriate or not; (2) using an inappropriate Broad Subject Area to narrow a topic, such as selecting "Arts and Humanities" to limit library user search scenario 5, which belonged to the "Physical Science/Engineering" subject area; (3) unable to use the limiting options implied in the predefined search scenarios. For example, library user search scenario 2 specified full-text, but many library user group subjects selected "Bibliographic databases" criteria option.

For Library Staff Group

Keyword Misused. The library staff group tended to use broader keywords to search the Selector, such as "psychology" instead of "consumer behavior" for staff search scenario 1. This often resulted in low recall searches. The library staff group was generally not doing well in finding best databases for library staff search scenarios 6 and 12. They seemed to have problems identifying the best keywords to search for these two questions. For example, some of the library staff group chose to use the author's name instead of the title of the play for search scenario 6 for the library staff group. For search question 12, the proper keywords were "women" and "computer science." However, most of the library staff subjects tended to translate the topic into a broader subject and used keywords "women" and "education" or "career." This searching behavior seemed associated with the subjects database searching habits.

Possible Solutions. "The purpose of a comprehensive system evaluation . . . is not simply to determine the present system performance level but also to discover the needs to be done to upgrade the performance of the present system" (Lancaster, 1969). After in-depth study, we found that some of the problems described above could be avoided by revising and improving the Selector's search interface, and by collapsing configurations of some limiting criteria. It is important to have a search interface that provides multiple options to help focus the search, while simultaneously avoiding requirement for an excessive number of decisions by the user. Some users might not have the knowledge to make the appropriate choices. Based on the analyses described above, we

decided to carry out the following system/interface improvements:

- (1) Collapsing "Database Level" limiting criteria options (from four selection choices into two). The "Database Level" picklist of the tested interface (see the Note of Fig. 1) included both academic levels and academic subject areas. These multiple options, though potentially useful for experienced users, were confusing for users unfamiliar with the broad subject areas. Moreover, these options provided opportunities for users to select Database Level criteria that conflicted with other limiting criteria. Halving the number of choices should reduce these problems.
- (2) Splitting the searching interface to distinguish between basic search level and advanced search level. A single searching interface with all optional criteria could be confusing for first-time users of the Selector. Having a two-level searching interface may help reduce the complexity and lower the learning curve for first-time and inexperienced users.
- (3) Eliminating the "Language" limiting criteria. The "Language" limiting criteria was originally included to help users distinguish databases that include only English publications from databases that include multiple language publications (including English). The usability test showed that many users applied this criterion in ways not anticipated.
- (4) Providing on-screen search hints and meaningful instructions.
- (5) Revising the keyword search form to accommodate various searching styles. The revised searching interface would allow one keyword search or nest-searching with phrases. It also automatically ignores stop-words entered by users.

Empirical Experience

We also found some disadvantages of using randomly selected subjects to perform usability tests with predefined searching scenarios:

- (1) Subjects lacked motivation or desire to find the information. They simply performed as required, not trying to understand the Selector and having little interest in evaluating the search results. Subjects played with the testing system out of curiosity, which contributed to many failed searches.
- (2) Subjects were not necessarily familiar with all of the topics or subject areas of the predefined search scenarios. The failed search analyses revealed that some of the subjects had problems identifying the appropriate keywords of the topics with which they were not familiar. Some of the subjects selected subject areas that contradicted the topic.

In addition to the usability test data described above, we also had empirical experience with the Selector. The Selector was installed on the Library's Central Information Desk and the Reference Desk in January 2000. Graduate Assistants and Library staff working at the Desk have been using

TABLE 10. Answers to the posttest question 1.

Question 1: I found that the Smart Database Selector was very useful in recommending appropriate databases for topics or term papers.											
Answers:											
Participant population	Strongly agree		Agree		Neither agree nor disagree		Disagree		Strongly disagree		Total subjects
	#:	%:	#:	%:	#:	%:	#:	%:	#:	%:	
Undergraduates	1		6		1						8
Graduates/faculty	1		5				1		1		8
Library staff	1		3		1		1				6
Total	3	13.6	14	63.6	2	9.0	2	9.0	1	4.5	22

the Selector regularly when they have difficulty finding a database/databases to search for a specific topic. We have also selectively let some library users use the Selector.

We observed differences between real users and usability test subjects using the Selector. Real users, when using the Selector to recommend electronic resources, would usually focus on keyword formulation and then moved on to the specific databases suggested. They were more interested in searching the suggested databases for relevant citations. On the contrary, usability test subjects were more interested in seeing how the Selector suggested different sets of electronic resources with different keywords and optional criteria. They paid little attention to the evaluation of the suggested electronic databases. We felt that some of the above failed searches could be avoidable if we were to associate the usability test with a class assignment.

User Satisfaction Measures

Satisfaction measures were based on the responses by all subjects to the posttest questionnaires, plus the brief interviews at the end of each test session. Tables 10 to 12 report the posttest responses to each of the user satisfaction-related questions. All measures were counted by actual number of

responses of the subjects. Percentages shown were calculated as percentages of subjects responding to the specific question. Overall, user reactions to the usefulness of the Selector were positive. Seventeen of the 22 subjects responding agreed that the Selector was useful, while two remained neutral and three did not deem it useful (see Table 10). Twenty of the 22 subjects would use the Selector alone or with the current Library menu system (see Table 11.). As for the search interface (see Table 12), 15 of the 22 subjects felt that the interface was easy or very easy to use while three remained neutral on this question and five disagreed. Undergraduate subjects appeared to like the Selector more than other groups. The undergraduate and library staff groups unanimously supported the idea to use the Selector alone or with the current Library menu system. Graduates and faculty subjects seemed to be able to navigate existing search interfaces better than the rest of the groups. Brief interviews after the test sessions revealed that those who did not support the use of the Selector tended to feel that they already know which resources to search for their research topic, and therefore did not need a computer to recommend databases to them. Tables 10 to 12 present the subjects' answers to the Post-Test Questionnaires.

TABLE 11. Answers to the posttest question 2.

Question 2: Compare the Smart Database Selector to the current library's static menu page. I would use:									
Answers:									
Participant population	This selector only		This selector and the current library menu		Not this selector		Other		Total subjects
	#:	%:	#:	%:	#:	%:	#:	%:	
Undergraduates	4		4						8
Graduates/faculty	2		4		1		1		8
Library staff			6						6
Total	6	27.2	14	63.6	1	4.5	1	4.5	22

TABLE 12. Answers to the posttest question 3.

Question 3: Overall, I found the Smart Database Selector's interface easy to use.											
Answers:											
Participant population	Strongly agree		Agree		Neither agree nor disagree		Disagree		Strongly disagree		Total subjects
	#:	%:	#:	%:	#:	%:	#:	%:	#:	%:	
Undergraduates	1		3		1		2		1		8
Graduates/faculty			7		1						8
Library staff			3		1		1		1		6
Total	1	4.5	14	63.6	3	13.6	3	13.6	2	9.0	22

Conclusions

The results of the test and evaluation were generally encouraging for the method used as a viable approach to facilitating end-user selection of databases. Data analysis showed that the average per question recall and precision rates were acceptable, although they varied from question to question and from search to search. The average per question precision rate varied from a high of 94.0% to a low of 10%. The majority of the per user recall rate for each search question against resources with controlled vocabulary (both real vocabulary and partial data-mined subject terms) were above 60%. As anticipated, the recall rate was much better when searches were done against database controlled vocabularies and partial data-mined subject terms than when such searches were done only against lists of subject category terms assigned to the electronic databases by librarians. We estimated that the recall and precision rates would be much better if we had all databases described with full controlled vocabulary terms and full-subject terms. The library staff group's recall rate was significantly higher than that of library user group, which indicates that search strategies and query formulation were important. User satisfaction measures showed that most library professional and library user test subjects felt confident in the usefulness of the Selector. We also reported on the different searching behaviors observed between randomly selected usability test subjects and real users. The test result showed that the Selector system could stand different searching styles from various user groups.

The test and evaluation process also helped us discover the specific areas that needed improvements and revisions. In the summer of 2000, we proceeded to modify the Selector. In the fall of 2000, with the above revisions in place, we performed a second usability test. The test results will be analyzed and published soon.

Future Vision

The research described has several unique facets. (1) We were able to analyze the database selection strategy used by reference librarians, and converted it into a database selec-

tion algorithm. The research revealed that it was possible to use this strategy in the computer-based database selection process to help decide and narrow the focus of the search to more relevant, user-preferred areas. This is of great value particularly given the current trend of content-sensitive, cross-database and cross-platform searching between Web resources. Although simultaneous access is helpful in allowing one single search query to access multiple Web resources, this technology also creates a new problem of "information/citation flooding." Our research showed the possibility of help to ease this problem by allowing users to limit their search query in a way that a professional librarian does in choosing databases to search. (2) This research made unique use of database-controlled vocabulary, and demonstrated the richness and usefulness of such terms even when searching using free-text end-user inputs. (3) Our research revealed the possibility that allowed a single free-text keyword search to simultaneously access and/or recommend relevant resource databases of different formats or media that were not only Web accessible, but also locally available, such as CD-ROM databases (see Ma, 2000). This has the advantage over other currently available commercial products, such as InfoTrac Total Access and WebFeat, which allow access to only Web accessible databases, and which require users to preselect a subject area, a restricted number of Web accessible databases, or to set up user profiles before proceeding to search.

Currently, updating and maintaining the Selector has been the topic of our continued research. There are two major parts of data included in the Selector tool: (1) database characteristic data and subject category terms; and (2) database thesauri/controlled vocabularies (see Ma, 2000). Updating is basically not necessary for database characteristic data and subject category terms. We found that these data tended to be stable over time. We compared database descriptions listed in the 1993 edition of *Gale Directory of Databases* with those in the 1999 edition. We observed relatively few changes in regard to general database characteristics for the databases covered in this prototype.

However, database resource thesauri/controlled vocabularies tend to be updated regularly to reflect new additions to

the databases and changing vocabulary in a given discipline. In this research project we were able to obtain database-specific controlled vocabularies from database producers for many of the databases characterized (both CD-ROM and Web databases). For some Web-accessible resources, we were also able to use data-mining techniques (relying primarily on Z39.50 protocols) to acquire samples of database-specific controlled vocabularies when the database vendor was unable to provide us with a complete database-specific thesaurus (see Ma, 2000). But, we did not establish systematic procedures for maintaining and updating database thesaurus information in this research project.

Our preliminary considerations regarding this issue are: Updating database thesauri-controlled vocabularies could be done either through downloading electronic thesauri or through data-mining. However, for either one of the methods, it is necessary to negotiate with individual producers or vendors for agreements and agreeable updating methods. It is common practice that many database producers routinely provide, to database subscribers, printed thesauri or attached on-line browseable thesauri for databases they produce to facilitate user access and searching of their products. For the purpose of developing and maintaining an electronic database advisory system as described in this study, it would be ideal if an agreement could be made between the subscribing institution and the database producer/vendor at the time of subscription agreement to provide electronic thesauri, or to set up a Web site or FTP site for routine downloading and updating of thesauri. For the library it would be necessary to establish an updating schedule, and to streamline the deleting and rebuilding of the indices process in a production implementation.

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